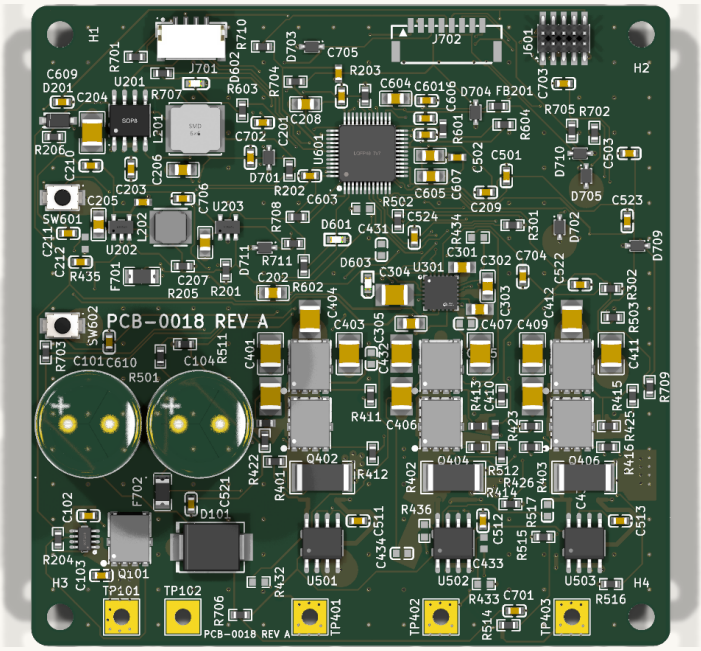
Owner's guide: 3-phase BLDC motor driver

What the board does, why its gate drive, current sensing, power loop and copper are built the way they are, how to wire it up, what is on it, and what still has to happen before anyone orders it.

PCB-0018-A is a 4-layer, 78 x 78 mm driver for a three-phase brushless (BLDC or PMSM) motor. It runs from a 10–28 V bus (12–24 V nominal) and is meant for 10 A RMS per phase continuous and 20 A peak, without a heatsink. Six 60 V N-channel FETs form three half-bridges, a DRV8300D gate driver switches them, three 3 mOhm shunts with INA240 amplifiers measure the phase currents, and an STM32G431 runs the control loop (sensored or sensorless field-oriented control). It is a 3-phase BLDC driver designed from a general spec. The only reference was a published demo board seen in one frame of a short video, and nobody published that board's spec or parts, so the spec here is ours.

All five build gates pass: ERC, placement, routed DRC, the design checks and DFM. The design checks pass with 390 waivers, explained below. Nothing has been built or measured, and nothing has been ordered. The 10 A RMS rating is a design target, not a proven one.



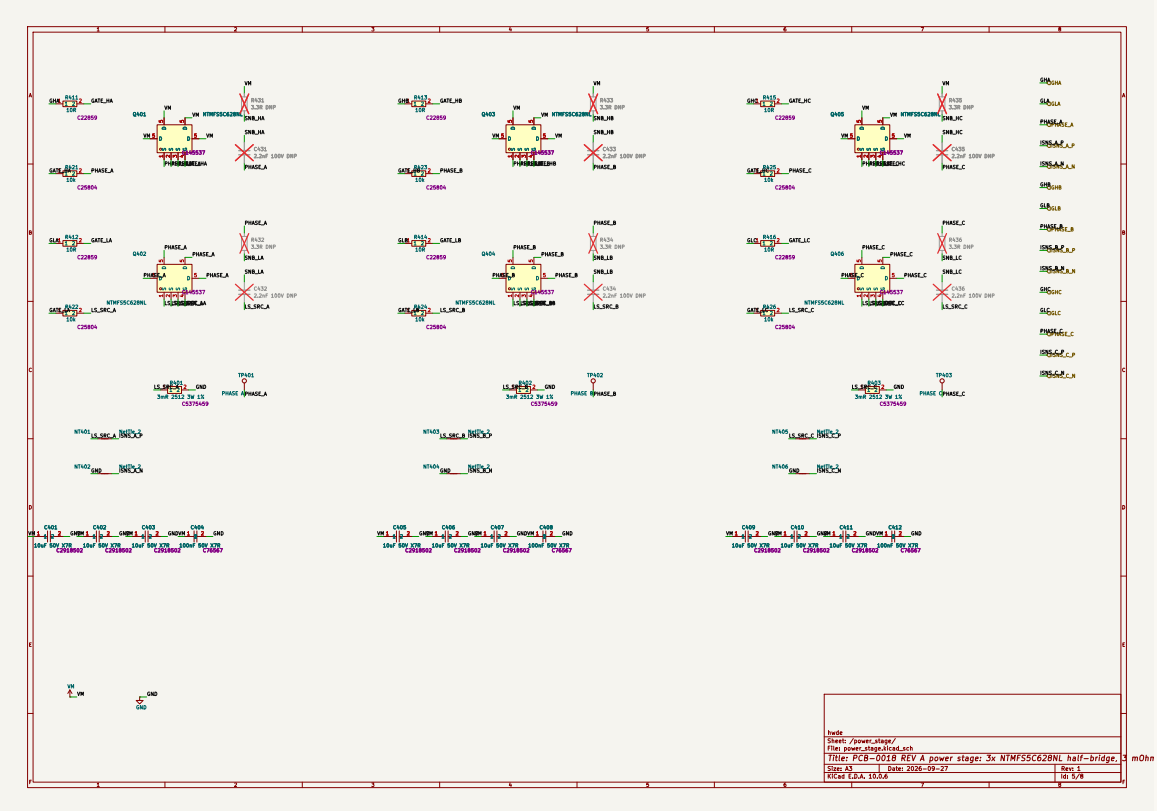
TOP SIDE, AS ROUTED 78 X 78 MM, 4 LAYERS, 2 OZ OUTER RENDER

The three half-bridges and their shunts sit in a row with the phase pads (TP401–TP403) on one edge and the power pads (TP101 VM, TP102 GND) beside them. The gate driver U301 sits just above the FET row, and the MCU, regulators and connectors fill the rest of the board.

Schematic

Eight sheets: power in, regulators, gate driver, power stage, sensing, MCU and I/O. The power stage is shown here.

FIGURE 1 *power stage sheet*



Three legs, each a high-side and a low-side NTMFS5C628NL, a 3 mOhm shunt from the low-side source to ground, per-leg ceramics and gate resistors. The other seven sheets are in `reports/schematic.pdf`.

Power comes in on TP101 (VM) and TP102 (GND) and passes a reverse-polarity FET (Q101, controlled by U101, an LM74500) before it reaches the bus, the SMCJ28A TVS and two 470 uF electrolytics. An LM5164 buck makes +5 V from the bus; a TPS61040 boost makes the +12 V gate supply from +5 V, and a TLV75733 LDO makes +3.3 V for the MCU and the current-sense

amplifiers. The MCU drives the gate driver with six PWM lines. Phase and bus voltages come back to the MCU through 150 k / 10 k dividers (1:16), so the 45 V clamp reads 2.8 V at the ADC.

How it works, and why it is built this way

Correctness came first, then size. These are the parts of the design that decide whether the board works.

Gate drive

The six power FETs (NTMFS5C628NL, 60 V, about 2 mOhm at 10 V gate drive, 5 x 6 mm) are driven by a TI DRV8300D, a three-phase half-bridge gate driver with its bootstrap diodes built in. The MCU drives it with six separate PWM lines (6-PWM mode), and dead time is set in hardware by the resistor on its DT pin (R301, 40.2 k), so a firmware mistake cannot shoot through a leg on its own.

The driver's gate supply, GVDD, does not come from the motor bus. The LM5164 buck makes +5 V from VM, and a small TPS61040 boost lifts that to +12 V for GVDD. That is the point of the design: the low-side gates see 12 V and the high-side gates about 11 V (12 V less the bootstrap diode drop and some droop) at every bus voltage from 10 V up to the TVS clamp. The requirement was at least 8 V of gate drive at VM = 10 V. We looked at the DRV8320H first, which makes its gate voltage with a charge pump from VM, but its datasheet guarantees only 6.3 V high-side drive at VM = 10 V, which fails that floor, so we dropped it.

Each high side runs from a 1 uF bootstrap capacitor (C301–C303) charged through the driver's internal diode whenever its low side is on. A bootstrap supply cannot hold a high side on forever, so the firmware must cap the modulation at about 95% duty and pulse the low sides once before the first high-side turn-on. Each gate has a 10 ohm series resistor (R411–R416) to slow the edges and damp ringing, and a 10 k pull-down to its source so the FETs stay off while the MCU is in reset or the +12 V rail is still coming up. The driver's own undervoltage lockouts on GVDD and on each bootstrap rail back that up. The DRV8300D has no drain-source overcurrent monitor, so overcurrent protection is the MCU's job: the current-sense outputs also feed the STM32G4's comparators, which are wired to the motor timer's break input.

Current sensing and the Kelvin connections

Each leg's low-side FET returns to ground through a 3 mOhm, 3 W, 2512 shunt (R401–R403). A 3 mOhm shunt drops only 30 mV at 10 A, so a few hundred microvolts picked up from the ground copper would be a visible error. For that reason the sense lines do not start from the ground pour. Each pair (ISNS_x_P and ISNS_x_N) starts at the shunt's own pads through a net tie, a deliberate short between two nets that KiCad lets us place exactly, so the amplifier measures the voltage across the resistor body and not across the copper that carries 10 A next to it. That is four-wire (Kelvin) sensing.

The amplifiers are INA240A1s, gain 20, with REF1 tied to +3.3 V and REF2 to ground, so the output sits at 1.65 V at zero current and moves 60 mV per amp. That gives about ± 27 A full scale, and the 20 A peak lands at ± 1.2 V around mid-scale. The INA240 was chosen over the cheaper INA181 because its reference split gives mid-supply with no extra parts and it tolerates a common-mode range of -4 V to 80 V, which matters when the shunt's ground end bounces during switching. A small RC filter sits at each amplifier's inputs.

One thing here did not come out as planned. The design checks want the two sense lines of a pair to run side by side, with no more than 3 mm apart. The INA240's two inputs sit on opposite rows of its package, 5.4 mm apart, and the shunt's two net ties are 3.46 mm apart, so each pair has roughly 9 mm where the lines cannot run together. The packages fix that length, not the routing. The six findings are waived, and the owner approved that waiver. The INA240's own PWM rejection and the low bandwidth of the sensed signal make this acceptable, in our judgement, but it is a judgement, not a measurement.

Power loop

The loop that matters most in a motor driver is the commutation loop: from the bus capacitors, through the high-side FET, the low-side FET and the shunt, and back to the capacitors. Every nanohenry in it turns into voltage overshoot on the switch node at each edge. Each leg therefore has its own three 10 uF / 50 V 1210 capacitors and a 100 nF placed straight across that leg, from the high-side drain to the shunt's ground end. The board has four layers: In1 is an unbroken ground plane under the whole power stage, and In2 carries VM under it, so the capacitors' return path runs directly under the loop.

The outer layers are 2 oz copper. The high-side drain pads have fields of 12–16 extra vias down to the VM plane, and the shunt ground returns have 12, 12 and 16 vias, after the design checks found the first layout dropping 10 A through only a few vias per leg. Two 470 uF / 63 V electrolytics hold the bus at low frequency, and an SMCJ28A TVS clamps transients at about 45 V. Reverse-polarity protection is a FET (Q101) driven by an LM74500 controller, so current can flow back into the supply during braking instead of pumping up the bus. There is no fuse on the board: the supply needs an external 20 A slow fuse.

The bulk capacitors are the weak point for continuous current. At 10 A RMS per phase about 5–6 A of ripple flows in the bus capacitors, most of it in the electrolytics at 20 kHz. Each 470 uF can is rated around 1–2 A of ripple (a guess; we could not get its datasheet), so 10 A continuous is more than the two cans are rated for. Running the PWM nearer 40 kHz moves more ripple into the ceramics.

Thermal

The board has no heatsink, so the copper is the heatsink. Our estimate at 10 A RMS per phase, 28 V and 25 kHz is about 3.4 W for the whole board: 1.8 W in the six stage FETs, 0.45 W in the shunts, 0.4 W in the reverse-polarity FET and 0.7 W in the regulators. A 4-layer, 2 oz board in still air sheds roughly 10–12 W per square metre per kelvin over both faces. At 78 x 78 mm that is about 7–8 C/W, so a mean rise of roughly 25 C with hotter spots at the FETs. That is a hand estimate scaled from TI's TIDA-010251 reference design, which measured 49 C at the FETs at 10 A RMS on a board about 60 x 60 mm. No thermal simulation or measurement has been done for this board, and the 10 A RMS rating stays provisional. Treat it as the target, not a proven rating.

Size

The board came out at 78 x 78 mm. The reference board looked roughly 60 x 60 mm in the video frame, but that is our guess from one frame. We put the gate loops and power loops first and let size follow: three legs of power copper, the reverse-polarity FET, a pair of 470 uF cans, a 100 V buck and a 12 V boost do not fit in 60 x 60 mm without crowding the gate loops. The extra area also helps the thermal estimate above.

What the checks waived, and what that means

ERC, placement, routed DRC, the design checks (verify) and DFM all pass. The design checks pass with 390 waivers, every one with its position and reason in `reports/verify-waivers.json` and the rule changes behind them in `routing/rules-added.md`. They split like this:

Check	Count	What they are
Current	359	282 via-count, 57 track-width and 20 pour-neck findings. Most (244) are on GND: stitching vias and small taps into the solid In1 plane that carry only their own part's return. The rest are Kelvin, gate-return, sense-divider and snubber taps on the phase and shunt nets, which carry signal current, and plane edges off the power path.
Creepage	24	7 inside the gate driver's (U301) courtyard and 5 inside the reverse-polarity controller's (U101) courtyard, where the package's own pin pitch sets the spacing (IPC-2221B Table 6-1, A6, component leads, 0.13 mm up to 100 V); 12 between nets of the same phase leg, which sit within about 15 V of each other. The owner set this scope on 2026-09-28.
Current-sense pairs	6	Skew and uncoupled length on ISNS_A/B/C, set by the INA240 and shunt pad geometry (above). Owner-approved.
Power distribution	1	The check reads GND as a supply with no decoupling. GND is the return, so this is an artifact of the check.

Placement has 14 more waivers of its own (`reports/place-waivers.json`): 12 where the placement check treats track-width rule areas as keepouts, and two courtyard overlaps that KiCad's own courtyard check passes.

One fix needs saying plainly. 109 short pad exits on GND, VM and the phase and shunt nets were narrower than the track-width rule for those nets. They were turned into solid copper zones of exactly the same shape, which the track-width rule does not measure. The copper did not get wider. Those necks are no better than before; they are just no longer counted. They are short pad exits, but if anything on this board runs hot first, look there.

What has not been done

Nothing has been built, powered or measured, and nothing has been ordered. Every current, temperature and ripple figure in this document is a calculation or a guess. The reverse-polarity FET, the bulk capacitors' ripple rating and the 10 A RMS continuous rating are the three things a first build should check.

Using it

Five wire pads for power and motor, two connectors, SWD, two buttons, three LEDs.

TABLE 1 *connections*

REF	CARRIES	NOTES
TP101, TP102	VM, GND	10–28 V DC. Fit an external 20 A slow fuse: the board has none. Reverse polarity is blocked by Q101.
TP401–TP403	Phase A, B, C	Motor windings, wire soldered or screw lug.
J701	GND, +5 V, UART TX, RX	JST-GH 4-pin. 3.3 V logic; +5 V out through a 0.2 A PTC.
J702	+5 V, GND, HA, HB, HC, ENC A/B/Z	8-pin sensor connector for Hall sensors or an ABZ encoder, 4.7 k pull-ups, +5 V through a 0.2 A PTC.
J601	SWD	2x5, 1.27 mm, the standard ST-Link / J-Link cable.
SW601, SW602	reset, user	
D601–D603	power, status, fault LEDs	

The board ships with no firmware. First power-up should be from a current-limited bench supply at 12 V with no motor connected: check +5 V, +12 V and +3.3 V, then program the MCU over SWD. The firmware must hold all six driver inputs low until +12 V is up, cap the PWM duty near 95% so the bootstrap capacitors recharge, and pulse the low sides before the first high-side turn-on. Set the comparator trip on the current-sense outputs before a motor is ever connected, since that is the only overcurrent protection. Then connect the motor and raise the current limit in steps while watching the FET and capacitor temperatures.

What's on it

148 machine-placed parts on 45 BOM lines: 17 Basic and 28 Extended unique parts. The power and phase pads and the four M3 holes are bare board features.

TABLE 2 *bill of materials*

DESIGNATORS	VALUE	LCSC	CLASS	QTY
C101,C104	470uF 63V	C106675	Extended	2
C102, ..., C610	100nF 50V X7R	C14663	Basic	16
C201, ..., C706	2.2nF 50V X7R	C1604	Basic	14
C202	22uF 25V X5R	C45783	Basic	1
C204, ..., C411	10uF 50V X7R	C2918502	Extended	11
C205	10uF 25V X5R	C15850	Basic	1
C206, ..., C607	1uF 50V X7R	C28323	Basic	9
C212	1nF 50V X7R	C1588	Basic	1
C404,C408,C412	100nF 50V X7R	C76567	Extended	3
C604	4.7uF 25V X5R	C1779	Basic	1
C606	10nF 50V X7R	C57112	Basic	1
D101	SMCJ28A	C2943858	Extended	1
D201	MBR0520L	C5563735	Extended	1
D601	LED green (power)	C12624	Extended	1
D602	LED yellow (status)	C2287	Extended	1
D603	LED red (fault)	C2286	Basic	1
D701, ..., D711	PESD5V0S1BA-ES	C5180216	Extended	8
F701,F702	PTC 0.2A hold 30V	C69680	Extended	2
FB201	600ohm@100MHz ferrite	C1002	Basic	1
J601	SWD 2x5 1.27mm	C2935458	Extended	1
J701	JST-GH 4p SMD	C189895	Extended	1
J702	JST-GH 8p SMD	C42376901	Extended	1
L201	22uH shielded	C2849539	Extended	1
L202	22uH shielded	C135264	Extended	1
Q101, ..., Q406	NTMFS5C628NL	C145537	Extended	7
R201	31.6k 1%	C25967	Extended	1
R202	33.2k 1%	C23003	Extended	1
R203, ..., R601	10k	C25804	Basic	14
R204	86.6k 1%	C23121	Extended	1
R206	100k	C25803	Basic	1
R301	40.2k 1%	C12447	Extended	1
R401,R402,R403	3mR 2512 3W 1%	C5375459	Extended	3
R411, ..., R416	10R	C22859	Basic	6
R501, ..., R711	100 ohm	C22775	Basic	11
R510,R512,R514,R516	150k	C22807	Basic	4
R602,R603,R604	1k 1%	C21190	Basic	3
R706,R707,R708	4.7k	C23162	Basic	3
SW601,SW602	Reset switch SMD	C720477	Extended	2
U101	LM74500-Q1	C5219210	Extended	1
U201	LM5164	C477928	Extended	1
U202	TPS61040	C7722	Extended	1
U203	TLV75733	C485517	Extended	1
U301	DRV8300DRGER	C3655801	Extended	1
U501,U502,U503	INA240A1	C2060769	Extended	3
U601	STM32G431CBT6	C529355	Extended	1

Class here follows the pre-buy list: every part on it is an Extended part and the rest are Basic. The full BOM, with MPNs and the do-not-place snubber footprints, is `fab/BOM-full.csv`.

What it costs

Not estimated yet.

No cost estimate was made for this board: the quote step was not run, so there is no `fab/quote.json`. With 28 unique Extended parts, JLC's per-part feeder fees will be a large share of a 5-board assembly order. Get the real number in JLC's own cart (`https://cart.jlcpcb.com/quote`) before ordering, or run `order_quote.py` first. The model spend of generating the board was not recorded either.

Ordering at JLCPCB

Written for a later day. Nothing here has been done, and nothing is ordered.

Before uploading anything, buy the Extended parts into JLC's inventory for 5 boards. JLC's BOM review can otherwise show one as idle stock, unselected at qty 0, until it has been bought in and the review re-run.

TABLE 3 *pre-buy list, 5 boards*

LCSC	PART	DESIGNATORS	PER BD	BUY
C106675	470uF 63V	C101,C104	2	10
C2918502	10uF 50V X7R	C204, ..., C411	11	55
C76567	100nF 50V X7R	C404,C408,C412	3	15
C2943858	SMCJ28A	D101	1	5
C5563735	MBR0520L	D201	1	5
C12624	LED green (power)	D601	1	5
C2287	LED yellow (status)	D602	1	5
C5180216	PESD5V0S1BA-ES	D701, ..., D711	8	40
C69680	PTC 0.2A hold 30V	F701,F702	2	10
C2935458	SWD 2x5 1.27mm	J601	1	5
C189895	JST-GH 4p SMD	J701	1	5
C42376901	JST-GH 8p SMD	J702	1	5
C2849539	22uH shielded	L201	1	5
C135264	22uH shielded	L202	1	5
C145537	NTMFS5C628NL	Q101, ..., Q406	7	35
C25967	31.6k 1%	R201	1	5
C23003	33.2k 1%	R202	1	5
C23121	86.6k 1%	R204	1	5
C12447	40.2k 1%	R301	1	5
C5375459	3mR 2512 3W 1%	R401,R402,R403	3	15
C720477	Reset switch SMD	SW601,SW602	2	10
C5219210	LM74500-Q1	U101	1	5
C477928	LM5164	U201	1	5
C7722	TPS61040	U202	1	5
C485517	TLV75733	U203	1	5
C3655801	DRV8300DRGER	U301	1	5
C2060769	INA240A1	U501,U502,U503	3	15
C529355	STM32G431CBT6	U601	1	5

Then, in order: (1) upload `PCB-0018-A_bldc-motor-driver_gerbers.zip`; (2) check the quote shows 4 layers, 78 x 78 mm, 1.6 mm and the JLC04162H-7628A stack-up with 2 oz outer copper; (3) enable PCB Assembly, top side; (4) upload `BOM.csv` and `CPL.csv`; (5) confirm every part matched its LCSC number, especially the Extended ones just bought in; (6) in the placement preview check by eye: the TVS D101 and the Schottky D201, the three LEDs, the two electrolytics C101 and C104, pin 1 of every IC, and the tab orientation of all seven FETs; (7) run JLCDFM as a second opinion; (8) pay. Payment is the owner's step, not a session's.